📘 MODULE 2: INSTALLATION + INTERFACE

2.1 Install on Ubuntu (reference)

You already have Splunk installed, so treat this as notes for your repo.

bash

sudo dpkg -i splunk-<version>-linux-amd64.deb

- Run in a terminal in the folder containing the downloaded .deb (from splunk.com, which needs a free account).
- dpkg -i installs a Debian package. Splunk lands in **/opt/splunk** (called $SPLUNK_HOME).
- Error "dependency problems": run sudo apt -f install.

bash

sudo /opt/splunk/bin/splunk start --accept-license

- First start only. It asks you to create the **admin username and password**.
- --accept-license skips the license prompt.

2.2 Start, stop, status

Command (run in terminal)	Meaning	Expected output
sudo /opt/splunk/bin/splunk start	Starts Splunk	The Splunk web interface is at http://host:8000
sudo /opt/splunk/bin/splunk stop	Stops it	Done
sudo /opt/splunk/bin/splunk restart	Stop + start (needed after most config changes)	Same as start
sudo /opt/splunk/bin/splunk status	Checks if running	splunkd is running (PID: ####)
sudo /opt/splunk/bin/splunk version	Shows version	Splunk 9.x.x (build …)

Breakdown of the pattern: sudo (admin rights) → /opt/splunk/bin/splunk (the CLI program) → a verb such as start.

Start on boot:

bash

sudo /opt/splunk/bin/splunk enable boot-start

Common errors

- *"Permission denied"*: you forgot sudo, or the files are owned by another user.
- *"Port 8000 already in use"*: another process holds the port. Splunk will ask to use a different one.
- *Browser can't connect*: Splunk is not running, or you typed the wrong port.

2.3 Default ports (interview favorite)

Port	Used for
8000	Splunk Web (browser)
8089	Management/REST port (splunkd)
9997	Receiving data from forwarders
8088	HTTP Event Collector (HEC)
514	Syslog (UDP/TCP, if you configure it)

2.4 Important directories (under /opt/splunk)

Path	Contains
bin/	The splunk CLI and other programs
etc/system/default/	Shipped default configs. **Never edit these.**
etc/system/local/	Your global config changes
etc/apps/	Installed apps and add-ons, each with its own default/ and local/
var/lib/splunk/	**Index data** (buckets)
var/log/splunk/	Splunk's own logs (splunkd.log, etc.)

2.5 Apps and add-ons

- **App:** has a user interface (dashboards, views). Example: Search & Reporting, Enterprise Security.
- **Add-on (TA):** no UI. Contains sourcetype definitions, field extractions, and inputs for a specific data type. Example: Splunk Add-on for Unix and Linux.

2.6 Splunk Web tour

Area	What it is
Search & Reporting	Where you run searches (your main workplace)
Settings	Indexes, data inputs, users/roles, lookups, knowledge objects
Apps menu	Switch apps
Monitoring Console	Settings → Monitoring Console. Splunk's health, indexing, and search performance
Activity / Messages	Jobs and system warnings

2.7 Hands-on lab: ingest a real log

Goal: bring /var/log/auth.log into Splunk.

1. Give Splunk permission to read it. Run in the terminal:

bash

```
ls -l /var/log/auth.log
```

ls -l shows owner and permissions. The file is usually root:adm, readable by group adm. Check which user Splunk runs as: ps aux | grep splunkd. If it's a splunk user, run:

bash

```
sudo usermod -aG adm splunk
```

(adds the splunk user to the adm group). Then restart Splunk. If Splunk runs as root, skip this.
2. In Splunk Web: **Settings → Add Data → Monitor → Files & Directories**, enter /var/log/auth.log, choose **Continuously Monitor**, **Next**.
3. Set the sourcetype (we'll learn why in Module 5), create/select index main, **Review → Submit**.
4. Search index=main source="/var/log/auth.log" with time **All time**.

Troubleshooting: If you see no events, it is usually a permissions problem. Check index=_internal source=*splunkd.log* ERROR.

2.8 Interview Q&A

- **Default Splunk Web port?** 8000. Management is 8089, receiving is 9997.
- **Which directory holds index data?** /opt/splunk/var/lib/splunk.
- **Why not edit system/default?** Updates overwrite it. Changes go in local.

- **App vs add-on?** An app has a UI, an add-on is data/config only.

2.9 Revision

$SPLUNK_HOME=/opt/splunk · start/stop/restart/status · ports 8000/8089/9997/8088 · never edit default · logs in var/log/splunk.